Epimorphisms, contractible mapping cones and surjections coincide for maps between sets

The Univalent Foundations Program

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Editorial note. The counted statement is the author's epis-surj (source lines 172-195); the proof environment begins two lines below it (line 196 is blank), so the pairing is by adjacency.

Editorial note. The proof invokes the dual of clause (iii) of the author's monomorphism lemma, so that lemma's statement and its proof are retained as context: the reference then resolves inside the package and no external reference has to be declared.

Editorial note. The retained counter values (chapter 10, section 1, theorem offset 0) reproduce the first-edition numbering: the retained monomorphism lemma prints as Lemma 10.1.1, the two intervening supporting lemmas keep 10.1.2 and 10.1.3, and the counted lemma prints as Lemma 10.1.4.

Editorial note. The statement's pushout diagram is retained exactly as drawn in the source (the retained preamble loads xy), and no figure file is needed.

Editorial note. No retained line refers to a bibliography entry, an included file, a label outside the retained ranges, or a non-ASCII character, so the package declares no external reference, no asset and no bibliography.

Editorial scope. One original result of the HoTT Book, counted once: for maps between sets, epimorphisms, contractible mapping cones and surjections coincide.

Allowed standard prerequisites (editorial).

- The mapping cone of a function between sets. For f : A -> B the mapping cone C_f is the pushout of f : A -> B along A -> 1, with constructors t : C_f, iota : B -> C_f and alpha_1 : prd{a:A} iota(f(a)) = t, presented in the statement by its pushout diagram.. The counted equivalence is stated between f being an epimorphism, C_f being contractible, and f being surjective; the proof runs induction on C_f (source lines 198-211).
- Categorical monomorphisms and their equivalent formulations. For f : hom_A(a,b) in a category A, being a monomorphism is equivalent to the contractibility of sm{h:hom_A(x,a)} (f o g = f o h) for all x and g (clause (iii) of book Lemma 10.1.1).. The proof applies the dual of that clause at source line 217 to obtain a path between the two pairs (iota, refl) and (const_t, alpha_1); the lemma and its proof are retained as context, so the reference resolves inside the package.

- Set-coequalizers and the universal property of the 0-truncated coequalizer. The coequalizer of f, g : A -> B in the category of sets is the 0-truncation of the homotopy coequalizer, and its universal property is the contractibility of sm{k:Q->C} (k o c = h).. Retained as part of the context because it fixes the categorical language in which the epimorphism side of the equivalence is read.

We now identify surjectivity and injectivity with the appropriate category-theoretic notions. First we observe that categorical monomorphisms and epimorphisms have a slightly stronger equivalent formulation.

Lemma 10.1.1. *For a morphism $f : \hom_A(a, b)$ in a category A, the following are equivalent.*

*(i) f is a **monomorphism**: for all $x : A$ and $g, h : \hom_A(x, a)$, if $f \circ g = f \circ h$ then $g = h$.*
(ii) (If A has pullbacks) the diagonal map $a \to a \times_b a$ is an isomorphism.
(iii) For all $x : A$ and $k : \hom_A(x, b)$, the type $\sum_{(h:\hom_A(x,a))}(k = f \circ h)$ is a mere proposition.
(iv) For all $x : A$ and $g : \hom_A(x, a)$, the type $\sum_{(h:\hom_A(x,a))}(f \circ g = f \circ h)$ is contractible.

Proof. The equivalence of conditions (i) and (ii) is standard category theory. Now consider the function $(f \circ -) : \hom_A(x, a) \to \hom_A(x, b)$ between sets. Condition (i) says that it is injective, while (iii) says that its fibers are mere propositions; hence they are equivalent. And (iii) implies (iv) by taking $k :\equiv f \circ g$ and recalling that an inhabited mere proposition is contractible. Finally, (iv) implies (i) since if $p : f \circ g = f \circ h$, then (g, refl) and (h, p) both inhabit the type in (iv), hence are equal and so $g = h$. □

Lemma 10.1.2. *A function $f : A \to B$ between sets is injective if and only if it is a monomorphism in $\mathcal{S}et$.*

Proof. Left to the reader. □

Of course, an **epimorphism** is a monomorphism in the opposite category. We now show that in $\mathcal{S}et$, the epimorphisms are precisely the surjections, and also precisely the coequalizers (regular epimorphisms).

The coequalizer of a pair of maps $f, g : A \to B$ in $\mathcal{S}et$ is defined as the 0-truncation of a general (homotopy) coequalizer. For clarity, we may call this the **set-coequalizer**. It is convenient to express its universal property as follows.

Lemma 10.1.3. *Let $f, g : A \to B$ be functions between sets A and B. The set-coequalizer $c_{f,g} : B \to Q$ has the property that, for any set C and any $h : B \to C$ with $h \circ f = h \circ g$, the type*

$$\sum_{k:Q\to C} (k \circ c_{f,g} = h)$$

is contractible.

Lemma 10.1.4. *For any function $f : A \to B$ between sets, the following are equivalent:*

(i) *f is an epimorphism.*

(ii) *Consider the pushout diagram*

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow & & \downarrow{\scriptstyle \iota} \\ \mathbf{1} & \xrightarrow[t]{} & C_f \end{array}$$

in $\mathcal{S}et$ defining the mapping cone. Then the type C_f is contractible.

(iii) *f is surjective.*

Proof. Let $f : A \to B$ be a function between sets, and suppose it to be an epimorphism; we show C_f is contractible. The constructor $\mathbf{1} \to C_f$ of C_f gives us an element $t : C_f$. We have to show that

$$\prod_{x:C_f} x = t.$$

Note that $x = t$ is a mere proposition, hence we can use induction on C_f. Of course when x is t we have $\mathsf{refl}_t : t = t$, so it suffices to find

$$I_0 : \prod_{b:B} \iota(b) = t$$
$$I_1 : \prod_{a:A} \alpha_1(a)^{-1} \cdot I_0(f(a)) = \mathsf{refl}_t$$

where $\iota : B \to C_f$ and $\alpha_1 : \prod_{(a:A)} \iota(f(a)) = t$ are the other constructors of C_f. Note that α_1 is a homotopy from $\iota \circ f$ to $\mathsf{const}_t \circ f$, so we find the elements

$$(\iota, \mathsf{refl}_{\iota \circ f}), (\mathsf{const}_t, \alpha_1) : \sum_{h:B \to C_f} \iota \circ f \sim h \circ f.$$

By the dual of Lemma 10.1.1(iv) (and function extensionality), there is a path

$$\gamma : (\iota, \mathsf{refl}_{\iota \circ f}) = (\mathsf{const}_t, \alpha_1).$$

Hence, we may define $I_0(b) :\equiv \mathsf{happly}(\mathsf{ap}_{\mathsf{pr}_1}(\gamma), b) : \iota(b) = t$. We also have

$$\mathsf{ap}_{\mathsf{pr}_2}(\gamma) : \mathsf{ap}_{\mathsf{pr}_1}(\gamma)_*(\mathsf{refl}_{\iota \circ f}) = \alpha_1.$$

This transport involves precomposition with f, which commutes with happly. Thus, from transport in path types we obtain $I_0(f(a)) = \alpha_1(a)$ for any $a : A$, which gives us I_1.

Now suppose C_f is contractible; we show f is surjective. We first construct a type family $P : C_f \to \mathsf{Prop}$ by recursion on C_f, which is valid since Prop is a set. On the point constructors, we define

$$P(t) :\equiv \mathbf{1}$$
$$P(\iota(b)) :\equiv \|\mathsf{fib}_f(b)\|.$$

To complete the construction of P, it remains to give a path $\|\mathsf{fib}_f(f(a))\| =_{\mathsf{Prop}} \mathbf{1}$ for all $a : A$. However, $\|\mathsf{fib}_f(f(a))\|$ is inhabited by $(f(a), \mathsf{refl}_{f(a)})$. Since it is

a mere proposition, this means it is contractible — and thus equivalent, hence equal, to $\mathbf{1}$. This completes the definition of P. Now, since C_f is assumed to be contractible, it follows that $P(x)$ is equivalent to $P(t)$ for any $x : C_f$. In particular, $P(\iota(b)) \equiv \|\mathsf{fib}_f(b)\|$ is equivalent to $P(t) \equiv \mathbf{1}$ for each $b : B$, and hence contractible. Thus, f is surjective.

Finally, suppose $f : A \to B$ to be surjective, and consider a set C and two functions $g, h : B \to C$ with the property that $g \circ f = h \circ f$. Since f is assumed to be surjective, for all $b : B$ the type $\|\mathsf{fib}_f(b)\|$ is contractible. Thus we have the following equivalences:

$$\begin{aligned}
\prod_{b:B} (g(b) = h(b)) &\simeq \prod_{b:B} \Big(\|\mathsf{fib}_f(b)\| \to (g(b) = h(b)) \Big) \\
&\simeq \prod_{b:B} \Big(\mathsf{fib}_f(b) \to (g(b) = h(b)) \Big) \\
&\simeq \prod_{(b:B)} \prod_{(a:A)} \prod_{(p:f(a)=b)} g(b) = h(b) \\
&\simeq \prod_{a:A} g(f(a)) = h(f(a))
\end{aligned}$$

using on the second line the fact that $g(b) = h(b)$ is a mere proposition, since C is a set. But by assumption, there is an element of the latter type. □